

PENCODER: Design for Prospective Memory and Older Adults

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ABSTRACT

The number of older adults with Mild Cognitive Impairment (MCI) is rapidly increasing. Individuals with this condition suffer from reduced daily activities and capacities, the resulting sense of frustration, and a decline in mental and physical health. However, there are not many existing studies for this specific user group in the HCI community. This study focused on addressing issues related to aging and memory in the MCI population. We designed a new interface, PENCODER, which supports prospective memory. The “design thinking process” and multidisciplinary human-centered perspectives were used in creating this interface. Findings and insights from this exploratory human-centered design approach suggest that it is imperative to conduct further research in HCI fields on (1) tools and technology features that support prospective memory, (2) measurements that assess the effect of using these kinds of tools, and (3) longitudinal perspectives for assessing self-determination outcomes in everyday interactions.

CCS CONCEPTS

• **Human-centered computing** → **HCI design and evaluation methods**.

KEYWORDS

User Experience, Design Thinking Process, Human-centered Design, Prospective Memory, Mild Cognitive Impairment (MCI)

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1 INTRODUCTION

Mild Cognitive Impairment (MCI) is a medical condition associated with loss of memory, reduced autonomy, and increased agitation [10]. It is one of the most common health complications of older adults [9]. Although the ambiguous nature of this disease makes diagnosing it challenging, it is estimated that sixteen million people struggle with MCI in the United States [1]. One symptom of MCI is a loss of prospective memory, defined as “the process or skills required to support the fulfillment of an intention to perform a specific action in the future [6, 20].” The loss of prospective memory is a primary factor in the decreased amount of daily activities, increased frustration, and decline in mental and physical health experienced by MCI patients. Prospective memory loss can cause disruptions to daily life, such as forgetting to close windows or doors, or even to turn off the stove [4]. For these reasons, interventions to support prospective memory could result in prolonged independence [4, 23], with possible ripple effects on self-esteem and life satisfaction [16].

Within the boundary of Behavioral Science and Human-Computer Interaction (HCI) fields, the last several decades have seen an increase in the understanding of prospective memory [17] as well as new technologies to support it [2]. Various attempts have been made to design technologies that support the well-being of the older population [13]. It is expected that new immersive technologies will contribute to extending the life span and enhancing the quality of life for this population within the context of daily human-computer interactions [25]. Especially when considering age-related memory loss, researchers and practitioners expect to tackle this issue by applying a variety of existing information and communication technologies [5]. However, much of this technology has not been specifically designed with the MCI population in mind, and there have been reports of steep learning curves and uncomfortable interfaces when trying to apply immersive technologies in the MCI context [4, 14].

To address these issues, the current study aimed to design a new type of prospective memory aid for older adults with MCI to support their existing lifestyle habits. We focused on this specific user group’s needs in relation to the use of technology in their established daily lives. We utilized the “design thinking process” approach, within the perspective of human-centered design, and conducted intensive group discussion sessions with design researchers and behavioral scientists. One of the outcomes of this process is an UX flow chart of a prospective memory aid for older adults with MCI to support designers in creating technologies for

the older population. We also designed a new smart-pen interface, which we named PENCODER. This device creates a seamless usage process that incorporates an alarm, calendar, and to-do list creation features into the user's physical writing process. We expect that this design study will facilitate future research on design for the older population with cognitive impairments.

2 BACKGROUND AND RELATED WORKS

Within the HCI and behavioral science literature, we primarily investigated existing solutions and technologies for older adults with overall prospective memory issues to discover novel design opportunities. As a result of the literature review with an exploration of existing tools and research findings, we identified three main categories of current design solutions and their limitations or challenges.

2.1 Memory-Training Tools

The first category of current design solutions to support prospective memory consists of memory-training tools [3, 19]. The majority of these solutions take the form of computer games or online applications. One of the limitations of this approach is ease-of-use, since the tools are available to individuals who have the necessary skills to operate them. These tools can present major obstructions for older adults, especially when the games are displayed on smaller screens that are not conducive to poor eyesight, or when it is easy to make mistakes in interacting with the technological devices [11, 16]. Furthermore, the memory-training approach requires using the product over a long period of time before any substantial benefits are evident, and it may incite frustration when progress is not immediately evident [3, 19].

2.2 Behavioral Habit Formation

The second category of design solutions focuses on behavioral habit formation. Positive behavioral intervention techniques (e.g., designing routines, implementing silent signals, assigning tasks, and setting expectations) are employed to create patterns within existing routines or tasks that will automatically prompt the individual to remember the next activity [21]. However, this approach is not much helpful for activities that occur out of the norm or in an unscheduled fashion. While scheduling can be an excellent way to support individuals with MCI in their daily routines, it does not assist with irregular items, which are the most likely to be missed or forgotten [4, 23].

2.3 Supporting Daily Activities

The third category of design solutions includes specific physical devices that are aimed at supporting the activities of people living with memory impairment. On the one hand, alarms and reminders linked to devices such as stoves can help to reduce the user's cognitive load and stress while preventing disasters such as fires. On the other hand, simple improvements such as automatic shut-off lights, faucets, and smart outlets can also alleviate memory burdens and save energy. In addition to merely compensating for memory lapses, tools in this category can actively prompt the user's prospective memory by encouraging them to recall an activity, event, or task

[12]. For example, a wide selection of pillboxes, some with alarms, can help users to remember their medication schedules more actively [7]. By providing additional cues (e.g., when an individual forgets what activity the alarm is for), the solutions in this third category can build prospective memory.

Although design suggestions for specifically dealing with the emerging needs of older adults with MCI are often not presented explicitly, a substantial body of relevant works reveals that greater ease-of-use and ease-of-access to technology have facilitated an increase in the number of tools in the market that specifically address prospective memory challenges [7, 18]. Specifically, we found that "writing out notes by hand in a daily context" has been found to support older adults' prospective memory [3]. Moreover, introducing notes, forming intentions, and creating plans when promoted by a digital memory aid were also associated with gain in prospective memory [7, 12, 15]. The research literature further indicated that digital memory aids were helpful in providing a backup prompt if an individual failed to remember an activity [7, 12, 24].

For these reasons, we decided to adopt the approach of memory support through a writing interface and to develop a user experience (UX) based on three strategies: (1) encoding thoughts and plans through handwritten notes, (2) planning out future activities, and (3) receiving reminders to improve the quality of daily life.

3 PRELIMINARY WORK AND DESIGN PROCESS

3.1 Design Methods

In the form of an exploratory case study to develop the design feasibility prototype for individuals with MCI, we used the "double diamond model," which is one of the well-known design thinking process approaches, over a period of eight weeks [22]. Through each of the four phases of the double diamond model – known as Discover, Define, Develop, and Deliver – we attempted to (1) discover core difficulties for adults in relation to MCI with prospective memory challenges, (2) define the target users' "pain points", (3) develop possible technological features and prototypes to address them, and (4) deliver and evaluate final human-centered solutions for the target group.

Along with twenty additional graduate students from design and behavioral science fields, the four authors participated in two hours regular workshop sessions every week. To gather divergent perspectives and refine our design findings, we organized and facilitated a group discussion in every workshop session, including focus group interviews with three older adults who are experiencing age-related memory loss (Discover), multiple users' need assessments (Define), ergonomic explorations of existing products and idea sketching (Develop), and paper prototyping with product storyboards (Deliver). As a result, this design thinking activity led to a final prototyping process involving multiple tools and solutions (Figure 1). Additionally, we considered that features included in our interface are widely accessible to the public to ensure that the solution is practical, attainable, and cost-efficient by reviewing the development process and evaluating the design and interface of the product.

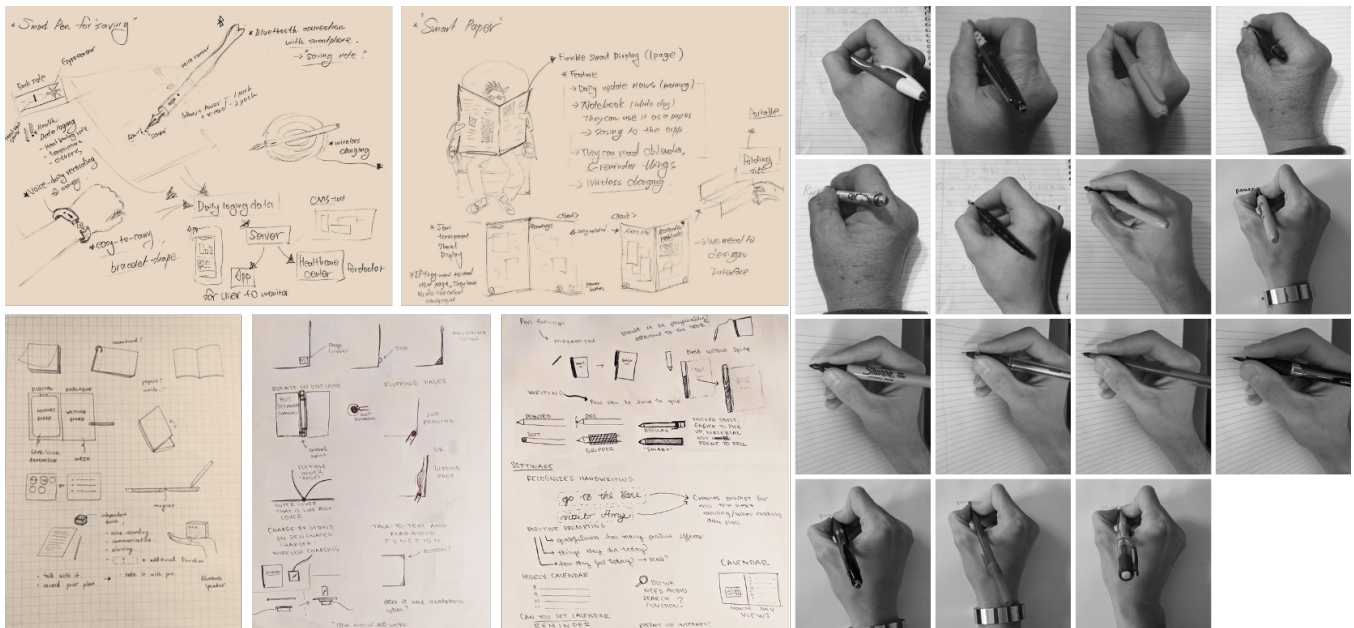


Figure 1: The Ideation Sketches and Mock-Up Testing (Left: Idea Sketch and Product Storyboards / Right: Ergonomics Research for Various Pen Grips) .

3.2 Design Results

3.2.1 User Study Results. Based on the three UX strategies for older adults with MCI that we identified from the relevant works and findings from our design thinking activities, we focused on the fact that more than 50% of memory problems that individuals face are related to prospective memory, and that these issues significantly impact everyday functioning [8]. Due to the pervasiveness of MCI in the older population, the design of a new tool could have long-standing impacts, such as an increased ability to live independently and a delay in the need for assisted living. This could in turn lead to economic advantages, both on an individual level and for overall society. The process of planning out activities, followed by reminders to complete them, may also increase the likelihood of engagement with family and friends, resulting in stronger social ties, particularly intergenerational ties [7].

We also noted that new tools in this area have the potential to propel the technological field forward, as their use could lead to more widespread familiarity with new technologies and a sense of technological competence [18]. This could have long-term implications for the growth of new technology applications in healthcare fields. Personal informatics derived from such tools can help inform health providers of broad patterns and to inform treatment plans. Such data can assist in adjusting technology settings for each user, while in its larger aggregated form it can help medical researchers better understand large-scale patterns of everyday activities and functioning in older adults. In this way, we expect that new technology specifically geared for adults with MCI, can positively impact not only the individual using the product but also society as a whole.

To provide design guidance for designers and developers on how to create new types of interfaces for older adults with MCI to support their existing lifestyle habits, we illustrated a UX flow chart

of a prospective memory aid for older adults with MCI (Figure 2). Our chart describes the overall UX flow and system features of a writing interface to support this specific target group by synthesizing and summarizing three UX strategies and an analysis of our user study findings.

3.2.2 UX and Interface Design. From the multi-layered background research on these academic findings and technological approaches, we decided to base the new technological interface on a traditional hand-held pen – an item that most older adults have been intimately acquainted with throughout their entire lives. With an understanding that the design needed to be accessible to less tech-savvy individuals, we focused on the ease-of-use of this new smart-pen interface. Various smart-pens are currently available to the public, but their features tend to be a bit complex and obscure, and they are not designed for or well-suited to older adults with MCI. Most centrally, existing smart-pens do not incorporate reminder features, which are critical for the needs of the MCI population.

To overcome the limitations of these extant solutions, we focused on how existing technologies and solutions can be adapted to this particular set of users and their core needs. The design of the PENCODER is intended to allow older adults with MCI by enabling them to pick up a familiar object and use it with no complex learning process. Wireless charging has been integrated into the pen so that older adults do not need to struggle with connecting a charging cord. Instead of connecting the charging cable, the unique wireless charger accompanying the pen is designed in the form of a traditional cup holder, allowing users to simply place the pen inside of the cup when not in use. In terms of use and ergonomics, the PENCODER strongly resembles more traditional pens, with special attention paid to its thickness, weight, and grip (Figure 3-a).

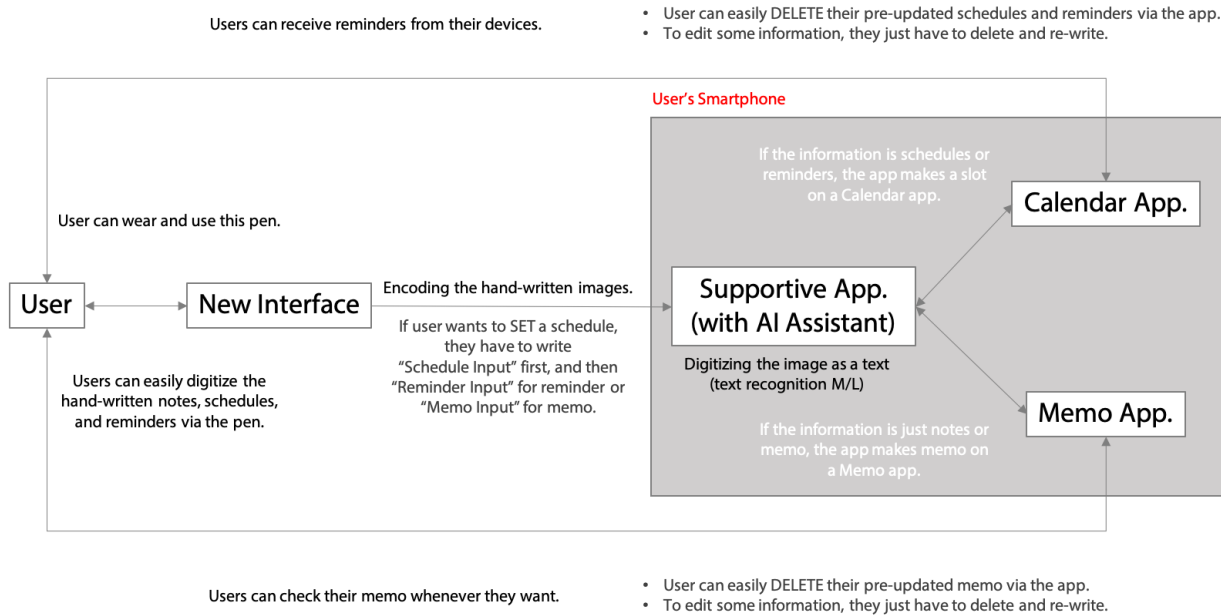


Figure 2: The Overall UX flow of the Prospective Memory Aid for Older Adults with MCI.

To support prospective memory, the PENCODER utilizes three separate technological features: (1) digitization of text through infrared cameras, (2) the automatic coding of items for specific lists and alarms on the PENCODER app (Figure 3-b), and (3) automatic speech recognition. The Infrared cameras on the pen allows the device to identify what is being written on a gridded paper, and thereby to digitize drawings and text. The text can be automatically converted to digital reminders, schedules, and to-do lists that can be checked and reviewed in the PENCODER app. By incorporating alarm, calendar, and to-do list creation into the writing process, the PENCODER eliminates the need to input data through a digital device. Moreover, the automatic speech recognition features, including a talk-to-text and read-out function, allow individuals to use the app and pen as usual even if their condition is not suitable for writing or reading [24] or if they are in a circumstance where it is not convenient (e.g., forgot the accompanying paper or left glasses at home). These features are helpful to the target user group because it provides a range of customization for each user, which is not currently provided by existing smart-pen technology. One of the greatest advantages of this approach is that it allows novice technology users and those familiar with advanced platforms to change the PENCODER settings to suit their needs. The goal is to create a seamless process that can gradually integrate advanced technological features into traditional note-writing without feeling foreign or new.

4 EXPERT EVALUATION WORKSHOP

4.1 Evaluation Methods

After the development of the PENCODER and the PENCODER app, we held a final design workshop to review the whole design process

and evaluate the feasibility of the interface for planning next steps. Twenty-seven reviewers, including four authors and twenty group discussion participants, participated in this workshop and shared feedback on the UX design concept and specific features within the lens of behavioral science and human-centered design. In this workshop, we also presented two video clips that illustrated the unmet needs of the MCI population (<https://youtu.be/9aTrC2pnykA>) and the advantages (<https://youtu.be/B0dcQWt7MhM>) that the PENCODER provides, along with our design thinking process and the ultimate technological outcome.

4.2 Evaluation Results

Most reviewers confirmed the value of the project and many praised its ability to inspire HCI researchers and practitioners to address the issue of aging and memory loss by utilizing human-centric perspectives and design thinking approaches within three aspects: (1) tools and technology features that support prospective memory, (2) measurements that assess the effect of using these kinds of tools, and (3) longitudinal perspectives for assessing self-determination outcomes in everyday interactions. Although this project focused on a very specific population, older adults with MCI, the reviewers also pointed out that this new interface can be replicated within the broader frameworks of UX strategy, resulting in outputs that could be beneficial for different user groups.

From a variety of informative ideas and design recommendations in the evaluation workshop, we also recognized that the next steps in developing and testing the PENCODER should involve additional needs assessments, followed by the creation of a demonstration prototype, and then in-depth user testing (Figure 4).

After fully functioning features have been implemented, this device will be tested with the target population to assess its

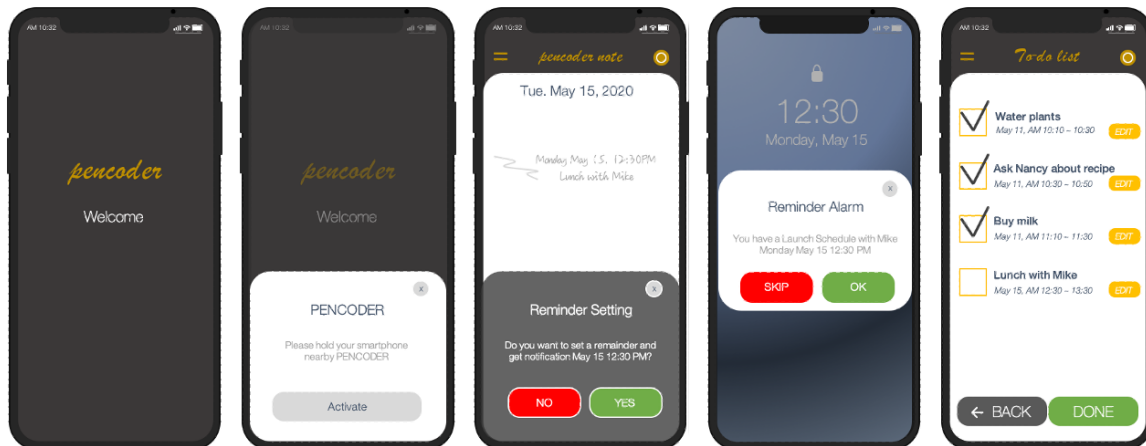
(a) PENCODER Device**(b) PENCODER App.**

Figure 3: Feasibility Prototype Examples of the PENCODER Device and PENCODER App.

ease-of-use and the timeline of product mastery. We expect that this step will entail at least some changes to the product based on user interviews in order to optimize the final outcome. Product testing will include the use of the pen by participants from the target group for a minimum of three months to allow sufficient time

for the device to become familiar and impact prospective memory. These experiments will be carried out with intermittent testing of the participants' memory capacities as well as interviews. During this testing the users will be able to explore the customization of the app settings if they are interested. If the product is found to be

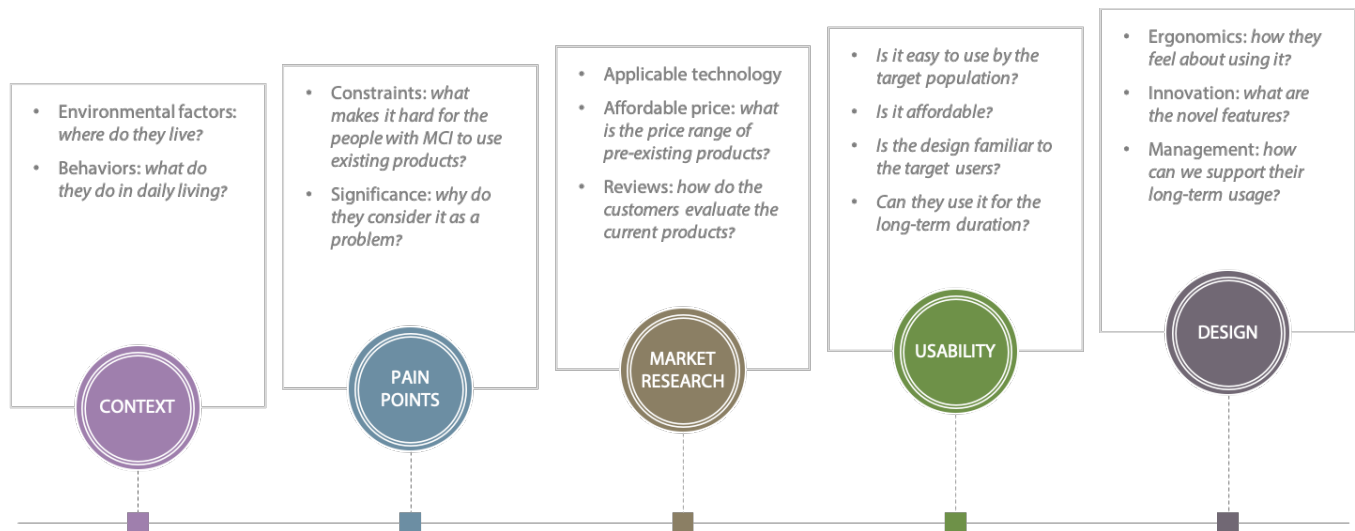


Figure 4: Need Assessment Questions Derived from This Study.

successful during the initial longitudinal tests, additional assessments will be made on self-determination outcomes through a combination of quantitative and qualitative measures.

5 CONCLUSION

This paper demonstrates an example of how the HCI community can create a new type of interface for a specific population using robust design techniques. By applying the design thinking model and developing the PENCODER on the basis of human-centered perspectives, we identified core “pain points” of the targeted user group as well as applicable UX strategies for future studies and design practices. There is a need for additional testing of the design outcomes, as well as further broad research on new tools and technological features to support prospective memory. Robust measures for assessing the long-term effect of using such tools also need to be developed and implemented. Taking lessons from our practical design case study, we believe that the implementation of the PENCODER can provide a foundation for the process of developing a variety of HCI features for older adults who would like to make each moment count.

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REFERENCES

- [1] Family Caregiver Alliance. 2020. *Incidence and prevalence of the major causes of brain impairment*. Family Caregiver Alliance. Retrieved February 24, 2021 from <https://www.caregiver.org/incidence-and-prevalence-major-causes-brain-impairment>.
- [2] Niamh Caprani, John Greaney, and Nicola Porter. 2006. A review of memory aid devices for an ageing population. *PsychNology Journal* 4, 3 (2006), 205–243.
- [3] Samantha WT Chan, Shardul Sapkota, Rebecca Mathews, Haimo Zhang, and Suranga Nanayakkara. 2020. Prompto: Investigating Receptivity to Prompts Based on Cognitive Load from Memory Training Conversational Agent. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 4, 4 (2020), 1–23.
- [4] Woong Choi. 2016. Design Approach to Support Prospective Memory for Independent Living among Aging People. *Archives of Design Research* 29, 1 (2016), 79–96.
- [5] Keith D Cicerone, Donna M Langenbahn, Cynthia Braden, James F Malec, Kathleen Kalmar, Michael Fraas, Thomas Felicetti, Linda Laatsch, J Preston Harley, Thomas Bergquist, et al. 2011. Evidence-based cognitive rehabilitation: updated review of the literature from 2003 through 2008. *Archives of physical medicine and rehabilitation* 92, 4 (2011), 519–530.
- [6] Judi Ellis and Lia Kvavilashvili. 2000. Prospective memory in 2000: Past, present, and future directions. *Applied Cognitive Psychology: The Official Journal of the Society for Applied Research in Memory and Cognition* 14, 7 (2000), S1–S9.
- [7] Scott Ferguson, Daniel Friedland, and Emma Woodberry. 2015. Smartphone technology: Gentle reminders of everyday tasks for those with prospective memory difficulties post-brain injury. *Brain injury* 29, 5 (2015), 583–591.
- [8] Brenda Hannon and Meredyth Daneman. 2007. Prospective memory: The relative effects of encoding, retrieval, and the match between encoding and retrieval. *Memory* 15, 5 (2007), 572–604.
- [9] Efraim Jaul and Jeremy Barron. 2017. Age-related diseases and clinical and public health implications for the 85 years old and over population. *Frontiers in public health* 5 (2017), 335.
- [10] Maria Johansson, Jan Marcusson, and Ewa Wressle. 2015. Cognitive impairment and its consequences in everyday life: experiences of people with mild cognitive impairment or mild dementia and their relatives. *International psychogeriatrics* 27, 6 (2015), 949–958.
- [11] Qingchuan Li and Yan Luximon. 2018. Understanding older adults’ post-adoption usage behavior and perceptions of mobile technology. *International Journal of Design* 12, 3 (2018), 93–110.
- [12] A McDonald, C Haslam, P Yates, B Gurr, G Leeder, and A Sayers. 2011. Google calendar: A new memory aid to compensate for prospective memory deficits following acquired brain injury. *Neuropsychological rehabilitation* 21, 6 (2011), 784–807.
- [13] Tracy L Mitzner, Julie B Boron, Cara Bailey Fausset, Anne E Adams, Neil Charness, Sara J Czaja, Katinka Dijkstra, Arthur D Fisk, Wendy A Rogers, and Joseph Sharit. 2010. Older adults talk technology: Technology usage and attitudes. *Computers in human behavior* 26, 6 (2010), 1710–1721.
- [14] Hazwani Mohd Mohadisudis and Nazlena Mohamad Ali. 2014. A study of smartphone usage and barriers among the elderly. In *2014 3rd International Conference on User Science and Engineering (i-USER)*. IEEE, New York, USA, 109–114.
- [15] Pam A Mueller and Daniel M Oppenheimer. 2014. The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological science* 25, 6 (2014), 1159–1168.
- [16] M Oriani, E Moniz-Cook, G Binetti, G Zanieri, GB Frisoni, C Geroldi, LP De Vreese, and Orazio Zanetti. 2003. An electronic memory aid to support prospective memory in patients in the early stages of Alzheimer’s disease: a pilot study. *Aging & Mental Health* 7, 1 (2003), 22–27.

- [17] Stephen J Payne. 1993. Memory for mental models of spatial descriptions: An episodic-construction-trace hypothesis. *Memory & Cognition* 21, 5 (1993), 591–603.
- [18] Dorian Peters, Rafael A Calvo, and Richard M Ryan. 2018. Designing for motivation, engagement and wellbeing in digital experience. *Frontiers in psychology* 9 (2018), 797.
- [19] Nathan S Rose, Peter G Rendell, Alexandra Hering, Matthias Kliegel, Gavin M Bidelman, and Fergus IM Craik. 2015. Cognitive and neural plasticity in older adults' prospective memory following training with the Virtual Week computer game. *Frontiers in Human Neuroscience* 9 (2015), 592.
- [20] Claire L Thompson, Julie D Henry, Adrienne Withall, Peter G Rendell, and Henry Brodaty. 2011. A naturalistic study of prospective memory function in MCI and dementia. *British Journal of Clinical Psychology* 50, 4 (2011), 425–434.
- [21] Robert Tobias. 2009. Changing behavior by memory aids: A social psychological model of prospective memory and habit development tested with dynamic field data. *Psychological review* 116, 2 (2009), 408.
- [22] Katja Tschimmel. 2012. Design Thinking as an effective Toolkit for Innovation. In *ISPIM Conference Proceedings*. The International Society for Professional Innovation Management (ISPIM), SOM I&O, Barcelona, Spain, 1.
- [23] Virginia G Wadley, Ozioma Okonkwo, Michael Crowe, David E Vance, Jennifer M Elgin, Karlene K Ball, and Cynthia Owsley. 2009. Mild cognitive impairment and everyday function: an investigation of driving performance. *Journal of geriatric Psychiatry and Neurology* 22, 2 (2009), 87–94.
- [24] Kiyoshi Yasuda, Tadashi Misu, Bobbie Beckman, Osamu Watanabe, Yoshinori Ozawa, and Tetsuo Nakamura. 2002. Use of an IC recorder as a voice output memory aid for patients with prospective memory impairment. *Neuropsychological Rehabilitation* 12, 2 (2002), 155–166.
- [25] Mary Zajicek. 2006. Aspects of HCI research for older people. *Universal Access in the Information Society* 5, 3 (2006), 279–286.